

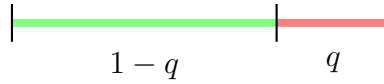
Geometric and Lattice Visualizations of Euler's Pentagonal Number Theorem Expansions

$$\prod_{k=1}^n (1 - q^k)$$

Part 1: Spatial Geometry ($n = 1$ to 3)

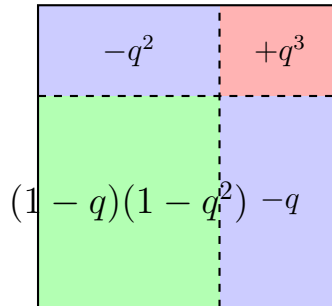
1D Geometry: $n = 1 \implies 1 - q$

A 1D line segment of length 1. We remove a segment of length q .



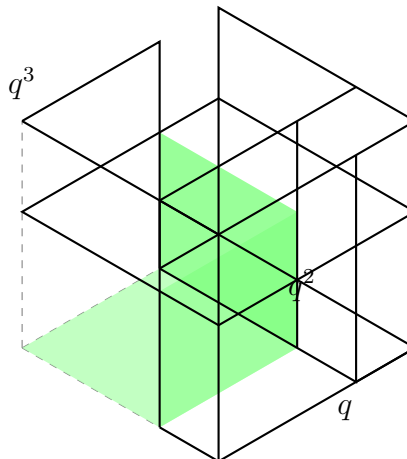
2D Geometry: $n = 2 \implies (1 - q)(1 - q^2) = 1 - q - q^2 + q^3$

A square where we remove a vertical strip q and a horizontal strip q^2 . The intersection q^3 is added back.



3D Geometry: $n = 3 \implies \prod_{k=1}^3 (1 - q^k) = 1 - q - q^2 + q^4 + q^5 - q^6$

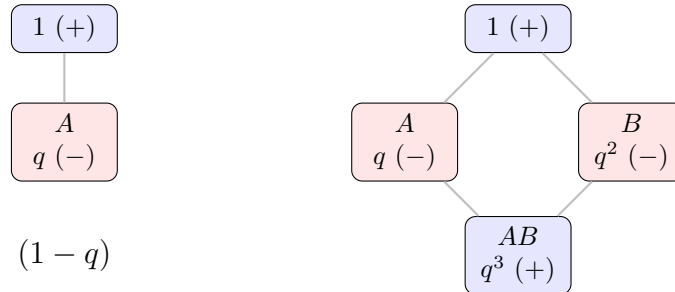
The intersection volume q^3 cancels identically with the third spatial slice q^3 .



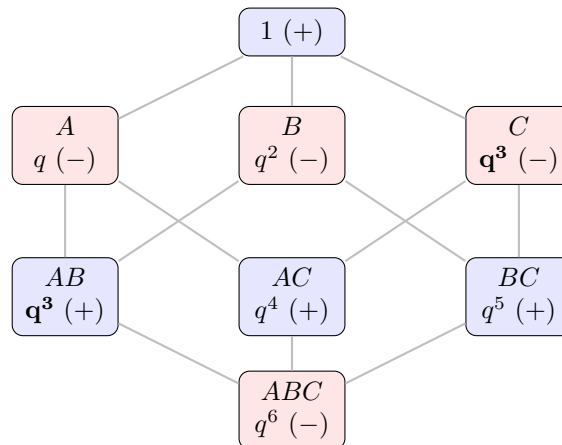
Part 2: Combinatorial Lattices ($n = 1$ to 5)

Blue nodes represent geometric additions (+). Red nodes represent subtractions (-).

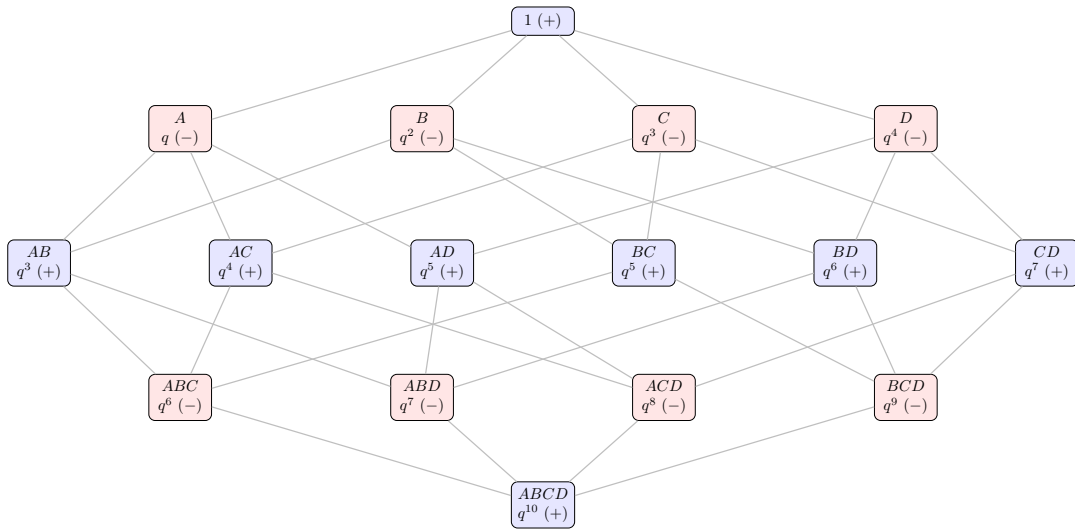
1 & 2. Lattices for $n = 1$ and $n = 2$



3. Lattice for $n = 3$: $1 - q - q^2 + q^4 + q^5 - q^6$



4. Lattice for $n = 4$: $1 - q - q^2 + 2q^5 - q^8 - q^9 + q^{10}$



5. Lattice for $n = 5$: $1 - q - q^2 + q^5 + q^6 + q^7 - q^8 - q^9 - q^{10} + q^{13} + q^{14} - q^{15}$

